U23AD491 – DATA SCIENCE

DS Sprint – Skill Based Assessment 2

Course Outcomes

CO3	Utilize data analytics techniques and perform predictive modelling for data-driven decision making.
CO4	Apply data pre-processing, EDA and feature engineering techniques to improve data quality and model performance.
CO5	Use Microsoft Excel and Python libraries (NumPy, Pandas, Matplotlib, Seaborn) to analyse data, create visualizations and derive meaningful insights.

Assessment Rubrics

Total Marks : 100

Round	Assessment Stage	Activities to Perform	Expected Output / Evidence	Marks
1	Quiz	• Answer multiple choice questions. • Questions from data analytics, pre-processing, EDA, feature engineering, Excel and Python libraries.	✓ Completed MCQ test. ✓ Score sheet.	20
2	EDA and Feature Engineering	• Load and understand the dataset. • Handle missing values and outliers. • Perform exploratory data analysis. • Create and encode new features. • Apply feature selection or dimensionality reduction.	Cleaned dataset. EDA visualizations (distribution plots, correlation, trends etc.). Engineered features. Feature selection or dimensionality reduction results.	30
3	Data Analysis Using Excel	• Clean and format data in Excel. • Use statistical functions. • Create charts and graphs. • Prepare Pivot Tables and Pivot Charts.	Well organized Excel file. Statistical analysis results. Charts, graphs and Pivot Table/Pivot Chart.	30
4	Pitching Business Insights	• Present key findings from your analysis. • Explain important patterns, trends and insights. • Provide data driven recommendations.	Presentation slides. Clear explanation of analysis results. Business insights and recommendations.	20



Marks Distribution

Round	Assessment Stage	Marks	Percentage
1	Quiz	20	20%
2	EDA and Feature Engineering	30	30%
3	Data Analysis Using Excel	30	30%
4	Pitching Business Insights	20	20%
Total		**100**	**100%**

USE CASE 25 - MACHINE MAINTENANCE INSIGHT

Understand Equipment. Detect Warning Patterns. Plan Maintenance.

Organization	**MechaCare Industrial Systems**
Sector	**Industry**

Business Context

MechaCare monitors industrial machines and wants to understand maintenance needs using sensor, usage, alarm and service records.

Business Question

What operating patterns indicate machines that require maintenance attention?

Dataset Description

File	Key Fields
sensor.csv	timestamp, machine, vibration, temperature, current
machines.csv	type, age, installation date
usage.csv	runtime, load, cycles
alarms.csv	alarm type, timestamp
maintenance.csv	service date, action, downtime

Data Quality Conditions

- Missing sensor intervals
- Duplicate timestamps
- Inconsistent alarm labels
- Extreme sensor values
- Incomplete maintenance history

ROUND 1 - QUIZ | 20 MARKS

- Answer objective questions from data analytics, data pre-processing, EDA, feature engineering, Microsoft Excel, NumPy, Pandas, Matplotlib and Seaborn.

Expected Evidence: Completed MCQ test and score sheet.

ROUND 2 - EDA AND FEATURE ENGINEERING | 30 MARKS

- Load and understand the datasets.
- Check data types, missing values, duplicate records and unusual values.
- Clean and combine the data at a suitable analysis level.
- Perform descriptive analysis and create meaningful visualizations.
- Study important relationships, patterns and trends.
- Encode categorical variables where required.
- Create meaningful new features.
- Apply feature selection or dimensionality reduction when appropriate.

Use-Case-Specific Analytical Questions

- Which machines show abnormal sensor ranges?
- How do readings change before maintenance?
- Which alarms occur most often?
- Does load relate to downtime?
- Which machine types require more service?
- What patterns appear in repeated failures?

Expected Evidence: Cleaned dataset, EDA notebook, key visualizations, engineered features and selected features.

ROUND 3 - DATA ANALYSIS USING EXCEL | 30 MARKS

- Import the cleaned data into Microsoft Excel.
- Organize and format the data clearly.
- Use suitable formulas and statistical functions.
- Create charts that answer the business questions.
- Prepare Pivot Tables and Pivot Charts.
- Create one summary sheet containing the important results.

Minimum Excel Analysis

- Sensor distributions
- Before-service trend
- Alarm frequency
- Load and downtime
- Machine-type comparison
- Repeat-failure patterns

Expected Evidence: Well-organized Excel workbook with formulas, statistical analysis, charts, Pivot Tables, Pivot Charts and a summary sheet.

ROUND 4 - PITCHING BUSINESS INSIGHTS | 20 MARKS

- Present the most important findings from the analysis.
- Support each finding with evidence from the data.
- Explain important patterns and their business meaning.
- Identify the areas that need management attention.
- Provide clear data-driven recommendations.
- Deliver a short, well-structured business pitch.

Pitch Focus

Recommend maintenance priorities for machines showing persistent warning patterns or repeated downtime.

Expected Evidence: Presentation slides, clear explanation of analysis results, business insights and recommendations.

Final Deliverables

- Cleaned and prepared dataset.
- Python notebook containing EDA and feature engineering.
- Excel workbook containing analysis and visualizations.
- Short business-insight presentation.

Marks Distribution

Round	Assessment Stage	Marks
1	Quiz	20
2	EDA and Feature Engineering	30
3	Data Analysis Using Excel	30
4	Pitching Business Insights	20
	Total	**100**